

Operator Discovery Report: restart_controller_two_phase_bottleneck_v1

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic restart-count controller with perturbation-enabled restarts; triggers more frequent restarts on stagnation (especially for two-cluster bottlenecks) while keeping growth bounded for transfer stability.
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:3:9:2.
- Surviving candidate: False.
- Problem class where it helps: clustered_tsp, two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 1c751f172211314c.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Runtime inflation is high at 1.51959.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 286.928971, distance-eval delta 10403.428571.
- cheapest_insertion_2opt gap delta -0.001632, runtime delta 951.0058, distance-eval delta 8589.142857.
- random_restart_2opt gap delta 0.0, runtime delta -151.159771, distance-eval delta -7988.0.
- sparse_three_opt gap delta 0.036775, runtime delta 59.4863, distance-eval delta 3681.571429.
- clustered_local_search gap delta -0.008943, runtime delta 8.116329, distance-eval delta 246.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.
- clustered_tsp operator gap 0.083393 vs baseline 0.110853.

- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.104566 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.005462.
- Runtime inflation: 1.51959.
- Distance-evaluation inflation: 1.664774.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.005462.
- fixed_restart_budget mean gap 0.115856 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.115856 and delta vs full operator 0.0.